Predicting Chronic Absenteeism via Socioeconomic Factors in Nebraska K-12 Schools

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Course Name / Capstone Title: DSCI/STAT 8950 Data Science Capstone Project
Date: April 24, 2026

Abstract

Chronic absenteeism is one of the most persistent and under-addressed problems in K-12 education across the United States. Nebraska schools have been particularly affected, especially in the years following COVID-19, with average rates nearly doubling from 8.5% in 2018-19 to a peak of 20.1% in 2021-22. This project investigates whether socioeconomic indicators, specifically Free and Reduced Lunch (FRL) eligibility, can reliably predict chronic absenteeism rates across Nebraska K-12 schools over a seven-year period from 2018 to 2025.

Using publicly available data from the Nebraska Department of Education, a panel dataset of 8,444 rows covering 1,298 schools and an ML-ready dataset of 361 rows across 73 matched schools in 10 counties were built. Ten machine learning models were trained and evaluated. Following professor feedback to use CV R-squared for model selection and CV RMSE versus Test RMSE for overfitting detection, the evaluation framework was updated accordingly. XGBoost emerged as the best model with a Test R-squared of 0.8143, Test RMSE of 0.0389, and a CV R-squared of 0.6441. FRL-related features combined account for 33.4% of feature importance, confirming poverty as the primary driver of absenteeism. The study estimates $15.8 million in education funding at risk over 7 years and classifies 23% of study schools as high-risk.

1. Introduction

1.1 Background and Motivation

Chronic absenteeism, defined as missing 10% or more of school days in an academic year, affects approximately one in six students in the United States. In Nebraska, the average chronic absenteeism rate across our study schools was 14.9% over the seven-year study period, with some schools reaching as high as 48.3% in a single year. The COVID-19 pandemic made things significantly worse, with rates nearly doubling between 2018-19 and 2021-22, and even by 2024-25 they remain 54% above pre-pandemic levels. This is not just an educational challenge but also a financial one, since school funding in the United States is tied to daily attendance.

Socioeconomic status is one of the strongest and most consistently documented predictors of educational outcomes. Students from low-income families face barriers to regular attendance including lack of transportation, food insecurity, health issues, and family responsibilities. Free and Reduced Lunch (FRL) eligibility is one of the most widely available school-level poverty

proxies, making it an ideal predictor for this type of analysis. If we can predict which schools are likely to have high chronic absenteeism before the school year ends, administrators and policymakers can intervene early rather than reacting after the fact.

1.2 Problem Statement and Objectives

The central research question for this project is: Can socioeconomic indicators, primarily Free and Reduced Lunch eligibility, reliably predict chronic absenteeism rates across Nebraska K-12 schools over a seven-year period? The specific objectives of this study are:

1. To build a clean, merged dataset from multiple NDE data sources spanning 7 years across 14 raw files.
2. To perform exploratory data analysis to understand trends, patterns, and the FRL-absenteeism relationship.
3. To engineer meaningful features and train 10 machine learning models from simple baselines to advanced ensembles.
4. To evaluate model performance using CV R-squared for model selection and CV RMSE versus Test RMSE for overfitting detection, following professor feedback.
5. To quantify the financial impact of chronic absenteeism and classify schools into risk tiers.

1.3 Scope of the Project

This study focuses on school-level analysis across 73 Nebraska schools in 10 counties over 7 academic years from 2018-19 through 2024-25. It is part of a larger two-scale capstone project where a partner analysis was conducted at the district level across 244 Nebraska districts. This individual report covers only the school-level component.

1.4 Structure of the Report

The remainder of this report is organized as follows. Section 2 reviews related literature. Section 3 describes the data sources and preprocessing steps. Section 4 presents exploratory data analysis. Section 5 explains the methodology and models used. Section 6 presents results and evaluation. Section 7 discusses the findings and challenges. Section 8 covers data ethics. Section 9 concludes with a summary and future directions. References and appendices follow.

2. Literature Review

2.1 Chronic Absenteeism and Its Consequences

Chronic absenteeism has been studied extensively in the education research literature. Attendance Works (2015) identifies it as one of the strongest early warning indicators of academic failure and dropout risk. Research by Chang and Romero (2008) found that children who are chronically absent in kindergarten and first grade are significantly less likely to read proficiently by third grade. At the high school level, Balfanz and Byrnes (2012) showed that chronically absent students are four times more likely to drop out than their peers. The financial consequences are also significant. In most U.S. states, school funding is tied to average daily attendance, meaning every absent student results in lost revenue for the district.

2.2 Socioeconomic Predictors of Absenteeism

Multiple studies have identified poverty as one of the strongest predictors of chronic absenteeism. Morrissey, Hutchison, and Winsler (2014) found that children from low-income families were significantly more likely to be chronically absent, with effects persisting even after controlling for family structure and parental education. The Free and Reduced Lunch program, which serves households earning below 185% of the federal poverty line, has been widely used as a school-level poverty proxy in education research (Harwell & LeBeau, 2010). In Nebraska specifically, rural counties with higher FRL rates have consistently shown higher chronic absenteeism rates.

2.3 Machine Learning in Education Research

The application of machine learning to education data is a growing area of research. Yadav et al. (2012) demonstrated that decision tree models could predict student performance with reasonable accuracy. More recent work by Bowers and Zhou (2019) applied gradient boosting models to early warning systems for dropout prediction, achieving strong results. XGBoost in particular has shown strong performance in education data contexts due to its ability to handle mixed data types, missing values, and small-to-medium sized datasets (Chen & Guestrin, 2016). The importance of proper model validation using cross-validation rather than a single train-test split for model selection is well established in the machine learning literature (Hastie, Tibshirani, & Friedman, 2009).

2.4 Gaps Addressed by This Study

Most existing work on absenteeism prediction focuses on individual student-level data, which raises significant privacy concerns. This study addresses that gap by working entirely with school-level aggregated public data. Additionally, the seven-year longitudinal panel design captures temporal trends including the COVID-19 structural break, which most earlier studies predate. The use of CV R-squared for model selection and CV RMSE versus Test RMSE for overfitting detection provides a more rigorous evaluation framework than is commonly reported in education machine learning studies.

3. Data Description

3.1 Source(s) of the Dataset(s)

All data for this project was sourced from the Nebraska Department of Education (NDE) public data portal. Three separate data sources were used:

6. Free and Reduced Lunch (FRL) counts by school: 7 annual files covering 2018-19 through 2024-25.
7. School Membership by Grade, Race, and Gender: 7 annual CSV files containing enrollment counts, school type, grade level, and county information.
8. Chronic Absence Reports: 7 annual district-level reports containing chronic absenteeism rates, which served as the target variable.

Data source: Nebraska Department of Education Data Center: https://www.education.ne.gov/

3.2 Key Features and Structure of the Data

After merging and preprocessing, two datasets were produced. The Panel Dataset contains 8,444 rows covering 1,298 unique schools across all 7 years. The ML Dataset contains 361 rows across 73 matched schools including the target variable CHRONIC_ABSENT_RATE.

Dataset	Rows	Schools	Years	Columns	Has Target
Panel Dataset	8,444	1,298	7	10	No
ML Dataset	361	73	7	13	Yes

The key features include FRL_PERCENT, FRL_COUNT, ENROLLED, TOTAL_STUDENTS, SCHOOL_TYPE, GRADE, COUNTY, and YEAR. The target variable is CHRONIC_ABSENT_RATE, a continuous value between 0 and 1.

3.3 Any Limitations or Assumptions About the Dataset

The primary limitation is that chronic absenteeism rates are available only at the district level, not the school level. All schools within the same district share the same target variable, which introduces some noise. Additionally, only 73 of the 1,298 panel schools could be matched to absence data, limiting the ML dataset size. It was assumed that district-level absenteeism rates are reasonable proxies for school-level rates within the same district.

3.4 Data Preprocessing Steps

The preprocessing pipeline involved several key steps. FRL files had headers buried in row 4 rather than row 0, requiring custom loading logic and fuzzy column matching. School ID formats were inconsistent across files, appearing in three different formats that were normalized before merging. The FRL and Membership datasets were merged on SCHOOL_ID and YEAR, and the chronic absence rate was merged using DIST_KEY. The final ML dataset had zero missing values after cleaning. Four missing values in the panel dataset were handled using median imputation within a scikit-learn Pipeline to prevent data leakage.

Second Level Headings

This report uses second-level headings throughout, following the Heading 2 style in MS Word as indicated in the template.

Third Level Headings

Third-level subsections follow this format using Heading 3 style in MS Word as directed in the template.

4. Exploratory Data Analysis (EDA)

4.1 Descriptive Statistics

The ML dataset contains 361 rows across 73 schools and 7 years. The chronic absenteeism rate ranges from a minimum of 3.57% to a maximum of 48.31%, with a mean of 14.99% and a standard deviation of 9.83%. School enrollment ranges from 25 to 1,189 students with a mean of 207.

Variable	Min	Max	Mean	Std Dev
CHRONIC_ABSENT_RATE	0.0357	0.4831	0.1499	0.0983
FRL_PERCENT	0.03	1.00	0.491	0.214
ENROLLED	25	1,189	207.3	208.9
TOTAL_STUDENTS	25	1,189	207.3	208.9

4.2 Data Visualizations

Thirteen charts were produced during EDA. Key visualizations include a bar chart of average absenteeism rate by year showing the COVID-19 spike, a scatter plot of FRL percentage versus chronic absenteeism rate with correlation line, a correlation heatmap of all numeric features, distribution histograms, box plots by school type and year, and county-level bar charts. All charts are available in the EDA notebook in the GitHub repository.

4.3 Insights Discovered Through EDA

Average absenteeism increased from 8.5% in 2018-19 to a peak of 20.1% in 2021-22 before declining to 13.1% in 2024-25, remaining 54% above pre-pandemic levels. The Pearson correlation coefficient between FRL_PERCENT and CHRONIC_ABSENT_RATE is $r = +0.42$, indicating a moderate positive relationship. Schools in higher FRL brackets consistently show higher absenteeism rates. Counties like BANNER, BOX BUTTE, and DAKOTA show both the highest FRL rates and the highest absenteeism rates.

4.4 Handling of Missing Data, Feature Selection and Engineering

The panel dataset had 4 missing values in FRL_PERCENT and FRL_COUNT, handled using median imputation within a scikit-learn Pipeline to prevent data leakage. The ML dataset had no missing values after cleaning. Seven new features were engineered: YEAR_NUM captures temporal trends, LOG_TOTAL_STUDENTS and LOG_ENROLLED reduce right skewness, FRL_PER_STUDENT captures poverty intensity, and SCHOOL_TYPE_ENC, COUNTY_ENC, GRADE_ENC are label-encoded categorical columns. The final feature set consisted of 11 features.

5. Methodology

5.1 Overview of Machine Learning Models Used

Ten machine learning models were trained and evaluated across three categories. Baseline models included Linear Regression, Ridge Regression, Lasso Regression, and Decision Tree. Linear Regression fits a straight line through the data. Ridge and Lasso add L2 and L1 regularization respectively. Decision Tree partitions the feature space through recursive binary splits. Ensemble models included Random Forest, Gradient Boosting, XGBoost in both default and GridSearchCV-tuned configurations, and LightGBM. These combine multiple models to capture non-linear relationships. A Multi-Layer Perceptron (MLP) neural network was also tested with L2 regularization and early stopping.

5.2 Novel Approaches Used

Several novel approaches were incorporated. First, a longitudinal seven-year panel design was used rather than a single-year snapshot, capturing temporal dynamics including the COVID-19 structural break. Second, FRL_PER_STUDENT was engineered as a poverty intensity ratio. Third, CV RMSE versus Test RMSE comparison was implemented as an overfitting detection framework following professor feedback. Fourth, school-level risk predictions were extended to all 1,298 Nebraska schools, enabling statewide risk mapping beyond the 73 study schools.

5.3 Rationale Behind Model Choices

The selection of 10 diverse models was intentional. Starting with simple linear models establishes a baseline and tests whether relationships are linear. Ensemble methods are well-suited to small-to-medium tabular datasets with mixed feature types. XGBoost was selected as a primary candidate because of its strong track record in education data contexts and ability to handle missing values natively. The neural network was included for completeness, to demonstrate that complexity alone does not guarantee better performance on small datasets.

5.4 Details of Train/Test Split, Cross-Validation, and Hyperparameter Tuning

The ML dataset was split into 271 rows for training (75%) and 68 rows for testing (25%) using random state 42. Following professor feedback, five-fold cross-validation on the training set was used as the primary model selection criterion through CV R-squared. CV RMSE was derived from CV R-squared using the formula: CV RMSE equals target_std multiplied by the square root of (1

minus CV R-squared), where target_std equals 0.0983. GridSearchCV with five-fold cross-validation was used to tune XGBoost, with a parameter grid covering learning rate, max depth, and n_estimators. The best parameters were learning rate = 0.01, max depth = 5, and n_estimators = 300.

5.5 Tools and Packages Used

The analysis was conducted entirely in Python using Google Colaboratory. Key packages include pandas and numpy for data manipulation, matplotlib and seaborn for visualization, scikit-learn for preprocessing and model training, xgboost and lightgbm for gradient boosting models, and geopandas for geographic mapping.

6. Results and Evaluation

6.1 Performance Metrics

Following professor feedback, the complete model comparison table below includes CV R-squared, CV RMSE, Test RMSE, Gap, and Verdict. Models are sorted by CV R-squared as the primary selection criterion.

Model	Train R2	CV R2	Test R2	CV RMSE	Test RMSE	MAE	Gap	Verdict
Random Forest	0.9688	0.7397	0.7130	0.0513	0.0483	0.0322	0.003	Good
Gradient Boosting	0.9378	0.7353	0.7667	0.0517	0.0436	0.0332	0.008	Overfit
LightGBM	0.9334	0.7101	0.7095	0.0541	0.0486	0.0349	0.006	Overfit
XGBoost Tuned	0.9426	0.6590	0.7753	0.0587	0.0428	0.0297	0.016	Overfit
XGBoost Default	0.9998	0.6441	0.8143	0.0600	0.0389	0.0249	0.021	Overfit
Decision Tree	1.0000	0.6010	0.4776	0.0635	0.0652	0.0342	-0.002	Good
Ridge Regression	0.2054	0.1273	0.1311	0.0939	0.0841	0.0632	0.010	Overfit
MLP Neural Net	0.8206	0.1266	0.4289	0.0940	0.0682	0.0537	0.026	Overfit
Linear Regression	0.2055	0.1241	0.1342	0.0941	0.0840	0.0631	0.010	Overfit
Lasso Regression	0.0000	-0.019	-0.004	0.1015	0.0904	0.0728	0.011	Overfit

Note: Gap equals CV RMSE minus Test RMSE. A positive gap means the model performs better on the held-out test set than during cross-validation, which is acceptable. A negative gap indicates genuine overfitting concern.

6.2 Visual Comparison

A scatter plot of actual versus predicted chronic absenteeism rates for the 68 test schools shows that most predictions cluster closely around the diagonal line, confirming model accuracy. Of the 68 test observations, 14 schools (approximately 20%) showed prediction errors above the RMSE threshold. A grouped bar chart comparing CV RMSE versus Test RMSE for all 10 models visually confirms which models have the largest gaps. All charts are available in the GitHub repository notebooks.

6.3 Discussion of Model Comparisons

The Decision Tree achieved a perfect Train R-squared of 1.0 but only 0.4776 on the test set, confirming it memorized all training examples. The MLP Neural Network performed worst with a Test R-squared of only 0.4289, confirming that 361 rows is insufficient for deep learning. XGBoost Default shows a gap of 0.021 but the direction is favorable since Test RMSE (0.0389) is lower than CV RMSE (0.0600), meaning the model actually performs better on the held-out test than during cross-validation. Random Forest has the best CV R-squared (0.7397) but its Test RMSE of 0.0483 is higher than XGBoost's 0.0389.

6.4 Final Model Recommendation and Justification

XGBoost Default is selected as the final model. It achieves the lowest Test RMSE of 0.0389 and the highest Test R-squared of 0.8143. While Random Forest has a higher CV R-squared (0.7397 versus 0.6441), it has a worse Test RMSE (0.0483 versus 0.0389), meaning XGBoost generalizes better to completely new schools. The gap direction of 0.021 for XGBoost is favorable, with the model performing better on the test set than during cross-validation. The Decision Tree and MLP are rejected due to perfect overfitting and insufficient data respectively.

6.5 Financial Impact Analysis

Using the formula: Chronically Absent Students multiplied by $12,500 per-pupil expenditure multiplied by 10% absence threshold, the total estimated financial impact across the 73 study schools over 7 years is $15.8 million.

Year	Financial Impact	Year	Financial Impact
2018-19	$1.17M	2022-23	$2.66M
2019-20	$2.06M	2023-24	$2.10M
2020-21	$2.69M	2024-25	$2.03M
2021-22	$3.10M (COVID Peak)	Total	$15.8M

6.6 Risk Classification

Schools were classified into three risk tiers. High Risk schools above 20% represent 23% of study schools. Medium Risk schools between 12 and 20% represent 23.9%. Low Risk schools below 12% represent 53.1%. When predictions were extended to all 1,298 Nebraska schools, 27.4% were High Risk, 19.3% Medium Risk, and 53.3% Low Risk.

7. Discussion

7.1 Interpretation of Results

The results strongly support the hypothesis that FRL percentage is a reliable predictor of chronic absenteeism in Nebraska K-12 schools. The positive correlation found in EDA ($r = 0.42$) is confirmed by the feature importance analysis, where FRL-related features together account for 33.4% of XGBoost's predictive power. The high importance of YEAR_NUM reflects the dramatic impact of COVID-19 on absenteeism patterns. The 2021-22 school year was so anomalous that the model essentially learned to treat it as a special case, reinforcing that intervention benchmarks should be set against pre-pandemic baselines.

7.2 How Well Does Your Model Solve the Problem?

The final XGBoost model explains 81.4% of the variance in chronic absenteeism rates with a Test RMSE of 0.0389, using only publicly available school-level aggregate data. This is a strong result given the limitations of working with school-level rather than student-level data and a relatively small dataset of 361 rows. On average, the model's predictions are within 3.89 percentage points

of the actual absenteeism rate, which is practically useful for identifying high-risk schools before end-of-year data is available.

7.3 Insights from the Modeling Process

Several important insights emerged. Simpler models like linear regression performed poorly (Test R-squared = 0.13), confirming that the relationship between poverty and absenteeism is non-linear. The Decision Tree's perfect training performance and poor test performance illustrates why train metrics alone are insufficient. Following professor feedback and implementing CV R-squared as the primary model selection criterion confirmed that the original choice of XGBoost was well-founded even under a more rigorous evaluation framework. The gap direction analysis added an important insight that a positive gap is not always a concern, since it means the model performs better on new data than during cross-validation.

7.4 Challenges Faced and How They Were Addressed

The most significant challenge was data preprocessing. The 14 raw files used three different ID formats, had headers in non-standard positions, and inconsistent column naming across years. This was addressed by building a robust custom preprocessing pipeline with fuzzy column matching and ID normalization. A second challenge was the limited ML dataset size, partially addressed by extending predictions to all 1,298 panel schools. A third challenge was the COVID-19 structural break, addressed by including YEAR_NUM as a feature allowing the model to learn year-specific patterns.

8. Data Ethics

Responsible data practices were maintained throughout this project. All data used is publicly available from the Nebraska Department of Education and contains no personally identifiable information. Analysis was conducted at the school and district level only, with no individual student records accessed or used at any point.

During EDA, visualizations were designed to be clear and non-misleading. Missing values were documented and handled transparently. All features were treated equally, and multiple models were compared to avoid cherry-picking the best result.

The use of FRL percentage as a predictor raises an important ethical consideration. FRL eligibility reflects systemic socioeconomic inequality, not individual choices. The model's findings should be interpreted as a call for systemic support and targeted resource allocation to high-need schools, not as a judgment on any school, community, or family.

This study was developed with awareness of real-world cases where data science caused harm. The COMPAS recidivism tool (2016) showed racial bias in criminal risk predictions. Cambridge Analytica (2018) violated informed consent by harvesting data from 87 million users. Amazon's AI hiring tool (2018) discriminated against women. These examples served as cautionary reminders to prioritize fairness and transparency throughout every stage of this project.

9. Conclusion

9.1 Summary of Findings

This study set out to answer whether socioeconomic indicators can reliably predict chronic absenteeism in Nebraska K-12 schools. The answer is clearly yes. XGBoost achieved a Test R-squared of 0.8143 and a Test RMSE of 0.0389, explaining 81.4% of the variation in chronic absenteeism rates. Following professor feedback, model selection was updated to use CV R-squared as the primary criterion and CV RMSE versus Test RMSE for overfitting detection. Even under this more rigorous framework, XGBoost Default remains the best model. FRL-related features account for 33.4% of feature importance, COVID-19 caused a near doubling of absenteeism rates, and $15.8 million in education funding is at risk over the study period.

9.2 Contributions of the Project

This project demonstrates that school-level absenteeism can be predicted with high accuracy using only publicly available aggregate data. The seven-year longitudinal design captures important temporal dynamics including the COVID structural break. The financial impact analysis and risk classification provide actionable outputs for Nebraska education administrators. The geographic analysis extends findings to all 93 Nebraska counties. The implementation of professor feedback on CV-based model selection adds methodological rigor not commonly reported in education machine learning studies.

9.3 Future Work or Improvements

Several directions for future work would strengthen these findings. A temporal train-test split training on earlier years and testing on later years would better simulate real-world deployment. SHAP values could provide per-school explanations of model predictions. Incorporating additional socioeconomic indicators beyond FRL such as median household income or housing stability could further improve accuracy. Expanding the financial model to the county level with actual district funding formulas would provide more precise cost estimates for policymakers.

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Appendices

Appendix A: GitHub Links for Code

All code, notebooks, and datasets for this project are available in the GitHub repository. The repository includes all three notebooks covering preprocessing, EDA, and machine learning modeling, along with the professor-feedback version of the ML notebook.

GitHub Repository: https://github.com/NileshPurnekar/Nebraska-Absenteeism-Capstone-2026

- preprocessing_&_merging.ipynb: Data loading, cleaning, and merging pipeline
- EDA.ipynb: Exploratory data analysis with 13 charts and visualizations

- Capstone_very_Fianl.ipynb: Original feature engineering, ML models, results (Steps 1-26)
- Capstone_very_Fianl_ipynb_Feedback.ipynb: Updated based on professor feedback, CV RMSE overfitting detection and CV R-squared model selection (Steps 27-30): https://github.com/NileshPurnekar/Nebraska-Absenteeism-Capstone-2026
- final_panel_dataset.csv: Cleaned panel dataset (8,444 rows)
- final_ML_dataset.csv: ML-ready dataset with target variable (361 rows)

Appendix B: Extra Charts and Visualizations

The following charts and visualizations were produced during this project and are available in the GitHub repository notebooks:

- Absenteeism trend bar chart (2018-19 through 2024-25) showing COVID-19 spike
- FRL% versus Absenteeism scatter plot with correlation line (r = 0.42)
- Correlation heatmap of all numeric features
- Distribution histograms for FRL_PERCENT and CHRONIC_ABSENT_RATE
- CV RMSE versus Test RMSE grouped bar chart for all 10 models
- Train R-squared versus Test R-squared comparison chart
- XGBoost feature importance horizontal bar chart
- Actual versus Predicted scatter plot with outliers highlighted
- Financial impact by year bar chart (2018-2025)
- School risk classification pie chart (73 study schools)
- Risk classification pie chart extended to all 1,298 Nebraska schools
- Nebraska school-level risk map
- Side-by-side Nebraska county comparison map: FRL% versus Predicted Absenteeism Risk

Appendix C: Detailed Model Output

Model	Train R2	CV R2	Test R2	CV RMSE	Test RMSE	MAE	Gap
Random Forest	0.9688	0.7397	0.7130	0.0513	0.0483	0.0322	0.003
Gradient Boosting	0.9378	0.7353	0.7667	0.0517	0.0436	0.0332	0.008
LightGBM	0.9334	0.7101	0.7095	0.0541	0.0486	0.0349	0.006
XGBoost Tuned	0.9426	0.6590	0.7753	0.0587	0.0428	0.0297	0.016
XGBoost Default	0.9998	0.6441	0.8143	0.0600	0.0389	0.0249	0.021

Decision Tree	1.0000	0.6010	0.4776	0.0635	0.0652	0.0342	-0.002
Ridge Regression	0.2054	0.1273	0.1311	0.0939	0.0841	0.0632	0.010
MLP Neural Net	0.8206	0.1266	0.4289	0.0940	0.0682	0.0537	0.026
Linear Regression	0.2055	0.1241	0.1342	0.0941	0.0840	0.0631	0.010
Lasso Regression	0.0000	-0.019	-0.004	0.1015	0.0904	0.0728	0.011

Appendix D: Meeting Logs and Team Responsibilities

This project was completed as part of a two-scale capstone study. Team collaboration and meeting logs are maintained on the course Microsoft Teams channel:

Microsoft Teams Channel: General | SP_2026_Team B: https://teams.microsoft.com

Team responsibilities were divided as follows:

- Nilesh Purnekar: School-level analysis covering 73 schools across 10 counties. Responsible for preprocessing, EDA, feature engineering, 10-model comparison, financial impact analysis, risk classification, and geographic mapping.
- Abhinav Adhikari: District-level analysis covering 244 Nebraska school districts. Responsible for district-level preprocessing, modeling, early intervention model, and district-level geographic analysis.
- Joint responsibilities: Combined presentation, joint EDA findings, shared data sources from Nebraska Department of Education, and final capstone presentation to faculty was presented on April 14, 2026.

Appendix E: Professor Feedback Implementation

The following changes were made in response to professor feedback received during the capstone course:

- CV R-squared was adopted as the primary metric for model selection in place of Test R-squared alone. This is implemented in Step 29 of the feedback notebook.
- CV RMSE versus Test RMSE comparison was added as an overfitting detection method. CV RMSE is calculated as target_std multiplied by the square root of (1 minus CV R-squared). This is implemented in Steps 27 and 28 of the feedback notebook.
- The complete model comparison table was updated to include CV R-squared, CV RMSE, Gap, and Verdict columns. This is implemented in Step 14b of the feedback notebook.

- Gap direction analysis was added to clarify that a positive gap is favorable, meaning the model performs better on the held-out test than during cross-validation. This is implemented in Step 30 of the feedback notebook.